

MPW HHI34 Customers

Executive Director
Prof. Dr. rer. nat. Martin Schell
Prof. Dr.-Ing. Thomas Wiegand

Einsteinufer 37
10587 Berlin

M. Sc. Axel Schönau
Project Manager – Photonic Components
Phone +49 30 31002- 494
axel.schoenau@hhi.fraunhofer.de
www.hhi.fraunhofer.de

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Multi Project Wafer Validation Report HHI34

Dear customers

This letter is to inform you about the wafer validation results of HHI34. Your chips will be shipped via FedEx and you will receive the tracking number in a separate message. The table below shows an overview of the validation results.

Waveguide loss	Butt-joint loss	PD+SSC responsivity	DFB wavelength deviation	DFB power at 50mA	DFB series resistance	SOA gain
1.8 dB/cm	1.2 dB	0.4 A/W	3 nm	2.2 dBm	7 Ω	11 dB

with best regards

Axel Schönau

Appendix A: Explanation of Measured Values and Wafer Validation Specification

Waveguide loss	Maximum loss over all polarizations and the C-band
Butt-joint loss	Loss of the active/passive transition over all polarizations, C-band
PD+SSC responsivity	Responsivity of a photodetector directly attached to a spot-size converter. Optical power is launched into the SSC via a butt standard single mode fiber. over all polarizations, C-band, butt-fiber
DFB wavelength deviation	Difference between DFB wavelength design target and the peak wavelength measured at 50mA. Positive values mean that the measured wavelength is redshifted
DFB power	Measured using a CW current of 50 mA on bar level without proper heat sink. The given value is the minimum power across the C-band. The measurement is done using an on-chip photodiode with known responsivity
DFB series resistance	Differential series resistance of a DFB laser above threshold
SOA gain	Gain of a 900 μm long SOA in TE. The given value is the minimum across the C-band and includes coupling loss of the active/passive transition but not any fiber coupling. The measurement is done with a CW current of 100 mA and without special heat management

Table 1: Explanation of measured values

component	parameter	typical value	pass criterium
waveguide			
	propagation loss (E200, E600 and E1700)	2 dB/cm	≤ 5 dB/cm
butt-joint			
	loss	2 dB	≤ 5 dB
photodetector			
	responsivity, including coupling via SSC	0.5 A/W	≥ 0.3 A/W
spot size converter			
	coupling loss to SSMF	1.5 dB	≤ 4 dB
DFB laser			
	power @ 50 mA	3 dBm	≥ 0 dBm
	wavelength accuracy @ 50 mA	$\leq \pm 4$ nm	$\leq \pm 8$ nm
	series resistance	12 Ω	≤ 20 Ω
SOA			
	gain @ 900 μm , 100 mA	10 dB	≥ 5 dB

Table 2: wafer validation specification

Appendix B: Detailed results

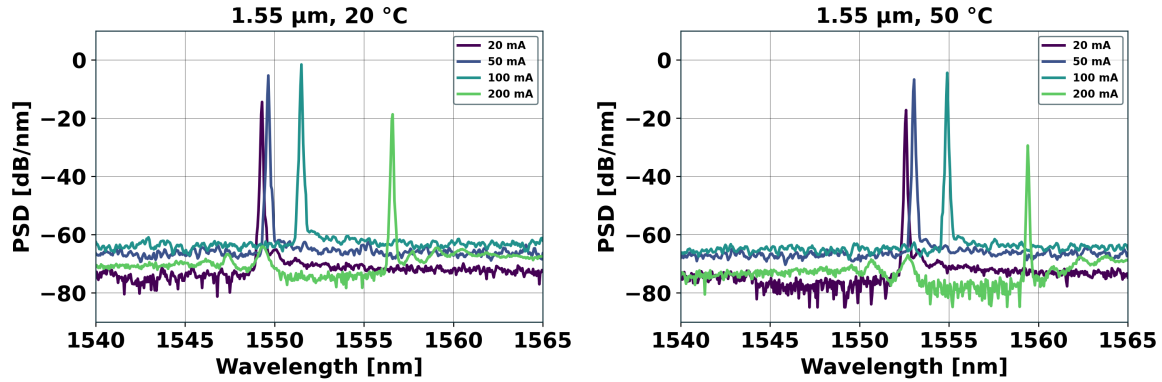


Figure 1: Spectrum of a DFB laser with a target wavelength of 1.55 μm, measured at various currents and at 20°C/50°C.

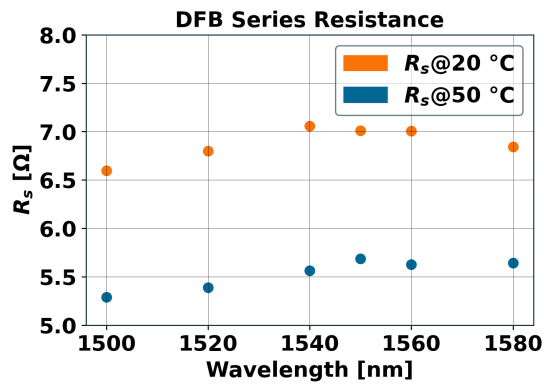


Figure 2: DFB series resistance above threshold for various DFB wavelengths at 20°C/50°C.

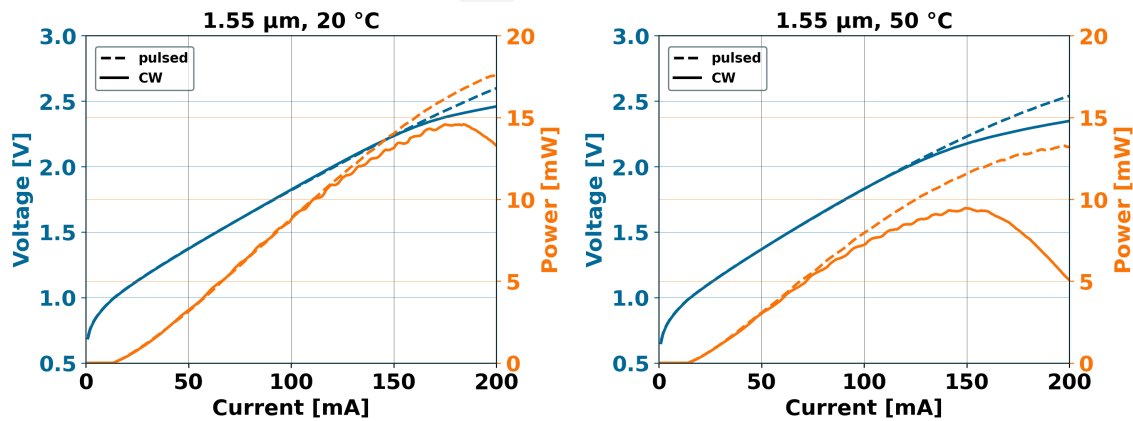


Figure 3: DFB LIV curves obtained using a CW current (solid) and a pulsed current (dashed) for 20°C and 50°C.